

Alyssa Hee

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EDUCATION

University of Toronto

Sept 2024 – Apr 2028

Bachelor of Applied Science in Electrical and Computer Engineering + PEY Co-op

CGPA: 3.77/4.00

- Intended Minor in Sustainable Energy
- **Relevant Courses:** Introductory Electronics, Digital Systems (Verilog), Programming Fundamentals (C++)

TECHNICAL SKILLS

Languages: HTML/CSS, JavaScript, C, C++, Java, Swift, Verilog

Frameworks: React, SwiftUI

CAD/Modelling: Altium, Fusion360, Blender

Developer Tools: Git, VS Code, Figma, Pixilart

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

University of Manitoba RIGA Lab

- Developing a Dual Active Bridge (DAB) converter running at 200kHz for use in a Solid State Transformer
- Designed model in PLECS and preparing to run Controller Hardware in the Loop simulation with completed DAB

Co-Webmaster

July 2025 – Aug 2025

F!rosh Week

- Redesigned the *orientation.skule.ca* website with a team of five using React, CSS, and MongoDB to match event branding and improve responsiveness across all device sizes
- Deployed site updates on remote servers via SSH, managing version control and updates through Git
- Provided user support for 1,000+ registrants, troubleshooting account issues, answering inquiries, and assisting with website dependent events like sign-in and Skule™ Hunt

SELECTED PROJECTS

Sparky PCB

June 2026 – Present

- Creating a PCB that can sense ESD events and toggle on an LED via a 555 Timer
- Simulated proposed circuit in LTSpice, successfully withstanding 35kV pulse modelled via HBM
- Routed 30mmx42mm PCB in Altium Designer, ensuring DRC was met. Assembly and testing coming soon

DE1-SoC Minicraft | [More info!](#)

Oct 2025 – Nov 2025

- Built a mini-Minecraft game in Verilog on an FPGA, using MIF-initialized ROM for tile maps to support real-time map updates
- Created gameplay logic including inventory counters, block placement/removal, and map initialization using modular FSMs and coordinate-based memory addressing
- Verified and debugged modular components in ModelSim before integration to resolve logic issues

EXTRACURRICULARS

Technical Team Co-Director

May 2026 – Present

University of Toronto IEEE Student Branch

Toronto, ON

- Running summer tutorials teaching PCB design from simulation to schematic and routing
- Developed and led an introductory workshop on Arduino, coordinating content and team collaboration to deliver an engaging hands-on learning experience
- Planned and co-lead a workshop on Altium for PCB design

Co-Captain

Sept 2025 – Present

SKULE A Intramural Mixed Volleyball

Toronto, ON

- Organized the team, inviting past players and recruiting new players at tryouts
- Coordinate player availability and scheduling, to ensure consistent attendance at games